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§ 2.4 A posteriori error analysis using the residual

After solving $Ax = b$ numerically, how can we gauge the effect of errors?

<u>Simple test</u> : $Ax = b$, $\hat{x}$ = computed solution

Define <u>residual</u> $\hat{r} = b - A\hat{x}$

Want $\hat{r}$ to be "small"

or

$\|\hat{r}\|$ is small

$\dfrac{\|\hat{r}\|}{\|b\|}$ is small

<u>Question</u> If $\|\hat{r}\|$ is tiny, is it always reassuring?

Let $\boxed{\delta b = -\hat{r}}$ Then $\boxed{A\hat{x} = b + \delta b}$

We know conditions that will guarantee

$$\hat{x} \approx x$$

when $\kappa(A)$ is reasonable.

Thus $\longrightarrow$

<u>Theorem</u> (restatement of "old" theorem)

Let $A \in M_n(\mathbb{C})$ be nonsingular, $b \neq 0$, $\hat{x}$ a computed approximation of x such that $Ax = b$. Let $\hat{r} = b - A\hat{x}$. Then

$$\frac{\|x - \hat{x}\|}{\|x\|} \leq \kappa(A) \frac{\|\hat{r}\|}{\|b\|} \qquad (*)$$

- We call $(*)$ an "a posteriori" estimate of the relative error (as opposed to "a priori") because it involves $\hat{x}$ & $\hat{r}$ that we've already computed.
- A posteriori estimates are typically used in methods where the computed solution will be "reused". (e.g. feed an iterative method for the solution of $Ax = b$).
- In fact, $\hat{x}$ can be any guess of x, or simply anything at all.

Chapter 3 The Least Squares Problem

Basic set-up

We have $y = p(t)$

p is nice (differentiable)

We don't know the function p

We can only observe $p = p(t)$ at a few points t_i, $i = 1, 2, \dots, n$

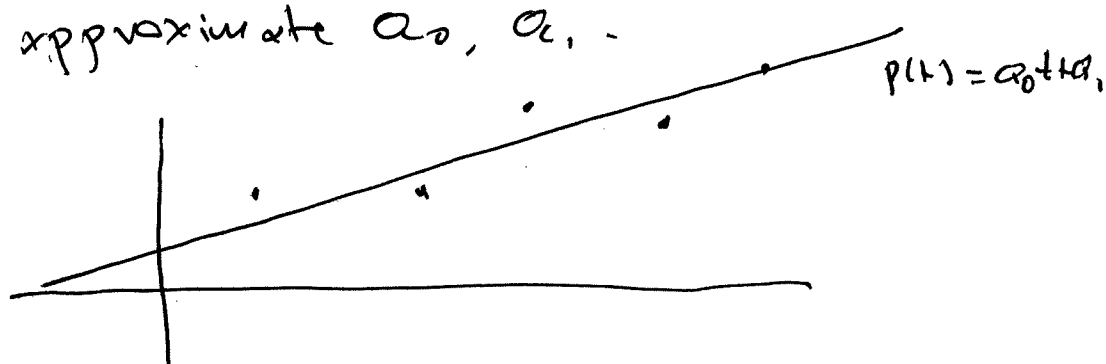
$$y_i = p(t_i), \quad i = 1, 2, \dots, n$$

We may have reasons to believe that $p(t)$ has a certain form, e.g.

$$p(t) \simeq a_0 t + a_1, \quad a_0, a_1 \in \mathbb{R}$$

(or more generally, a polynomial, an analytic function etc.)

We have a "optimization" method to approximate a_0, a_1.



Effort: find $p(t)$ so that the error in missing the known points is minimized.

To gauge how good such a line $p(t)$ can be in estimating data point $\forall t \in [a,b]$ we define an error to be minimized by our choice of $p_1(t)$ (a_0, a_1):

Consider $|y_i - p(t_i)|$, $i = 1, 2, \ldots, n$;

$$r_i := y_i - p(t_i)$$

$$r = (r_1, r_2, \ldots, r_n) \in \mathbb{R}^n$$

Minimize

$$\|r\|_2 = \left(\sum_{i=1}^{n} |y_i - p(t_i)|^2 \right)^{1/2}$$

The method of minimizing $\|r\|_2$ is called least squares method (LS) & the problem LS Problem (LSP), because λ^2 are involved.

<u>Example</u> (to illustrate the linear algebraic formulation of LSP)

known (t_i, y_i), $p(t_i) = y_i$, $i = 1, 2, \ldots, n$

Let $\{\phi_1, \phi_2, \ldots, \phi_m\}$ for the vector space of polynomials of degree $\leq m-1$

E.g. $\{1, t, t^2, \ldots, t^{m-1}\}$

[Typically: $n \gg m$, here $n = 100$, $m = 1$]

Then
$$p(t) = \sum_{j=1}^{m} x_j \phi_j(t)$$

So
$$p(t_i) = \sum x_j \phi_j(t_i) = y_i, \quad i = 1, 2, \ldots, n$$

But we can write this:

linear system!
$$\begin{bmatrix} \phi_1(t_1) & \phi_2(t_2) & \cdots & \phi_m(t_m) \\ \cdots & \cdots & \cdots & \cdots \\ \phi_1(t_n) & \phi_2(t_n) & \cdots & \phi_m(t_n) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

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The linear system to be solved depends on the right-hand side & the coefficient matrix; it may be inconsistent or consistent with multiple solution ($n >> m$). We need to select "solutions" in the least-squares sense (solve the LSP)

We need basic "inner product space" theory.

distances
orthogonality
angles

Quick review $\longrightarrow$

§ 3.2 Orthogonal Matrices, rotations, reflections

$\mathbb{R}^n$ (or $\mathbb{C}^n$) : view as an inner product space (ips), i.e. the usual vector space along with an <u>inner product</u>

$$\mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}$$

$$\langle x, y \rangle = \sum_{j=1}^{n} x_j y_j = x^T y = y^T x$$

$$\forall\, x, y \in \mathbb{R}^n, \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Basic properties of ips

- $\langle x, y \rangle = \langle y, x \rangle$
- $\langle x, \alpha_1 y + \alpha_2 z \rangle = \alpha_1 \langle x, y \rangle + \alpha_2 \langle x, z \rangle$
- $\langle \beta_1 x + \beta_2 y, z \rangle = \beta_1 \langle x, z \rangle + \beta_2 \langle y, z \rangle$
- $\langle x, x \rangle \geq 0$, $\langle x, x \rangle = 0 \iff x = 0$

(In $\mathbb{C}^n$, $\langle x, y \rangle = y^* x$, $\langle x, y \rangle = \overline{\langle y, x \rangle}$ etc.)

i.p.s's allow us to:

- Define <u>norm</u> of a vector $x \in \mathbb{R}^n$:

$$\|x\|_2 = \sqrt{\langle x, x\rangle} = \left(\sum_{j=1}^{n} x_j^2\right)^{1/2}$$

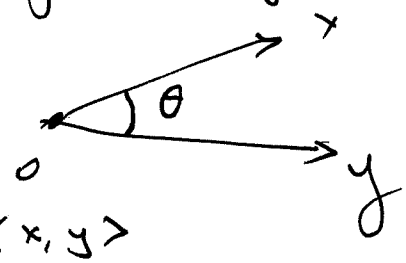
(length of a vector)

- $|\langle x, y\rangle| \le \|x\|_2 \|y\|_2$ (Cauchy-Schwartz)

- $\|x+y\|_2 \le \|x\|_2 + \|y\|_2$ (Triangle)

- Basic (analytic) geometry:

In $\mathbb{R}^2$



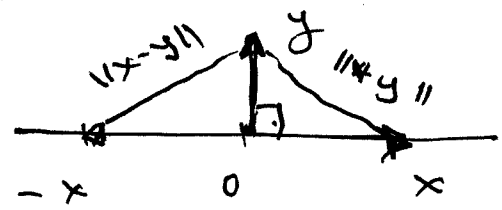
$$\cos\theta = \frac{\langle x, y\rangle}{\|x\|_2 \|y\|_2}$$

In $\mathbb{R}^n$ we'll use

$$\theta = \arccos \frac{\langle x, y\rangle}{\|x\|_2 \|y\|_2}$$

- $x, y \in \mathbb{R}^n$ are <u>orthogonal</u> $(x \perp y)$ if $\langle x, y\rangle = 0$

Justification in $\mathbb{R}^2$:



$x \perp y \iff \|x-y\| = \|x+y\|$

$\iff \langle x-y, x-y\rangle = \langle x+y, x+y\rangle$

$\iff \cdots \iff \langle x, y\rangle = 0$

Orthogonal matrices

Definition $Q \in M_n(\mathbb{R})$ is called orthogonal ($Q \in M_n(\mathbb{C})$ called unitary) if $Q^T Q = I$ $\quad (Q^* Q = I)$

(of course, equivalent to saying $QQ^T = I$, $Q^T = Q^{-1}$)

Fact Q is orthogonal, $Q = [q_1\ q_2 \cdots q_n]$ columns

Then $Q^T Q = \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix} [q_1\ q_2 \cdots q_n]$

$$= \begin{bmatrix} q_1^T q_1 & q_1^T q_2 & \cdots & q_1^T q_n \\ \cdots & \cdots & \cdots & \cdots \\ q_n^T q_1 & q_n^T q_2 & \cdots & q_n^T q_n \end{bmatrix}$$

$$\overset{\text{has}}{\underset{\text{to}}{=}} I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 \cdots & 0 \\ \cdots & & & \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$

So $\quad q_j^T q_j = 1 \quad , \; j = 1, 2, \ldots, n$

$q_i^T q_j = \langle q_j, q_i \rangle = 0 \quad$ if $i \neq j$

i.e. $\{ q_1, q_2, \ldots, q_n \}$ is an

<u>orthonormal set</u>

- vectors are of unit length:

$$1 = q_j^T q_j \Rightarrow \| q_j \|_2 = 1$$

- mutually orthogonal

$$\langle q_i, q_j \rangle = q_j^T q_i = 0 \; , \quad i \neq j$$

(Characterization of orthogonal matrices)

Basic properties of orthogonal matrices that will be useful in solving LSP:

1) $\forall x \in \mathbb{R}^n$ & orthogonal $Q \in M_n(\mathbb{R})$,

$$\|Qx\|_2^2 = \langle Qx, Qx \rangle = (Qx)^T Qx$$
$$= x^T Q^T Q x$$
$$= x^T I x$$
$$= x^T x = \|x\|_2^2$$

i.e. $\|Qx\|_2 = \|x\|_2$

Orthogonal matrices preserve length.

2) More generally.

$$\langle Qx, Qy \rangle = y^T Q^T Q x = y^T x = \langle x, y \rangle$$

3) As a consequence of 1) & 2)

In LSP we want to minimize over x

$$\|b - Ax\|_2$$

Equivalent to minimizing

$$\|Qb - QAx\|_2$$

We can use orthogonal matrices as useful preconditioners in solving $Ax = b$.

TBC